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Why Would Creatine Cause Insomnia?

This seems to be a rarely reported side effect, but common enough to address.

CHRIS MASTERJOHN, PHD
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In Creatine: Far More Than a Performance Enhancer, Alex Leaf and I discussed a conference abstract reporting insomnia as a side effect of creatine supplementation and two of my consulting clients who reported the same. We did not have a clear conclusion but suggested it was a transient response to cutting the methyl demand nearly in half.

On Consuming Creatine in Foods and Supplements, Arvind commented, “Creatine gives me insomnia. . . .”

After thinking about this more, I believe this is likely driven by inhibition of glycine synthesis. Glycine is an inhibitory neurotransmitter and 3 grams before bed has been shown to improve sleep.

Creatine will cut down the demand for S-adeonsyl-methionine (SAMe) by about 45%. The SAMe levels will rise, shutting off MTHFR. This will cause a backup of 5,10-methylene-tetrahydrofolate.

Most glycine is synthesized from serine, and most serine is derived from glucose.

But the conversion of serine to glycine is reversible. Converting serine to glycine requires tetrahydrofolate and produces 5,-10-methylene-tetrahydrofolate. Converting glycine to serine, by contrast, requires 5,-10-methylene-tetrahydrofolate and produces tetrahydrofolate.

Thus, an accumulation of 5,-10-methylene-tetrahydrofolate without MTHFR- and methionine synthase-mediated release of tetrahydrofolate will strongly inhibit the synthesis of glycine and will strongly favor the conversion of glycine to serine.

This is over and on top of any loss of glycine that occurs from overmethylating it to sarcosine and dimethylglycine, and this is not necessarily to the exclusion of other mechanisms that could produce insomnia, such as SAMe activating trans-sulfuration without adequate molybdenum and complex IV activity to clear the resulting sulfite, which itself enhances glutamate activity.

I would think the body could adapt to this by producing less SAMe over time, whereas this might be a more enduring side effect of actually supplementing with SAMe.

If this explanation is correct, comprehensive nutritional screening should show low glycine, high serine, and high formiminoglutamate (FIGlu).

However, if comprehensive screening for energy metabolism suggests the NADH/NAD+ ratio is elevated, this could mask the FIGlu by preventing the NAD+-dependent breakdown of histidine. This could potentially make the problem even worse, by increasing the NAD(P)+ transhydrogenase-dependent production of NADPH, favoring even greater production of 5,10-methylene-tetrahydrofolate. In this case, the low glycine and high serine would be the key signal.

The obvious solution would be to stop the creatine. However, if the creatine were essential for some purpose such as sports performance, the best way to mitigate this effect would likely to be 1) restricting the methionine content of the diet, 2) keeping high folate status, and 3) supplementing glycine.

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